UMHACKATHON 2026

StockWise

AI-Powered Inventory Optimizer for Malaysian SMEs

by Team Rojak

PROBLEM STATEMENT

The Problem

1. SMEs contribute **38% of Malaysia's GDP** but lack AI-powered inventory tools
2. Rule-based systems cannot analyze market news, exchange rates, and supply chain risks
3. RM/USD currency swings of **5-15% per quarter** make procurement timing critical
4. Manual spreadsheet tracking leads to stockouts, overstocking, and missed opportunities

OUR SOLUTION

Predictive Intelligence

Combines structured data (inventory, sales, exchange rates) with unstructured market intelligence (news)

Reasoning Chains

AI-powered reasoning chains explain WHY each recommendation is made

Market-Aware

Integrates real-time news, exchange rates, and supply chain signals

Strategy Simulation

Multi-scenario comparison for informed decision-making

FEATURES

Key Features

AI Recommendations

News-informed analysis with transparent reasoning chains and priority-ranked actions

Strategy Simulation

Compare Conservative, Balanced, and Aggressive scenarios with trade-off analysis

Real-time Dashboard

KPI tracking, sales velocity charts, and exchange rate monitoring

AI ARCHITECTURE

AI as a Reasoning Engine

Inventory + Sales + Exchange Rates + Market News

Structured Prompt (~2000 tokens)

Z AI GLM Analysis (z-ai-web-dev-sdk)

Prioritized Recommendations + Reasoning Chains

Not just alerts — Predictive, explainable intelligence

PLATFORM

8 Integrated Modules

Dashboard
Real-time KPI overview

Inventory
Full CRUD stock control

Products
Product & supplier mgmt

Sales Velocity
Product movement analytics

Exchange Rates
RM/USD monitoring

Market News
News feed & sentiment

AI Reco
LLM-powered insights

Strategy
Multi-scenario comparison

AI TRANSPARENCY

AI That Explains Its Thinking

6

Market News Articles

Analyzed per Generation

100%

Transparent Reasoning

Chains on Every Reco

3

Strategy Scenarios

Conservative/Balanced/Aggressive

TECHNOLOGY

Technology Stack

Frontend

- Next.js 16
- React 19
- Tailwind CSS 4
- shadcn/ui
- Recharts

Backend

- API Routes
- Prisma ORM
- SQLite
- Z AI GLM SDK
- Bun Runtime

IMPACT

Projected Business Impact

15-30%

Procurement Cost Savings

Research-backed

8

Core Feature Modules

Fully Operational

7

Database Models

Full CRUD Support

LIVE DEMO

See It Live

Full-stack web application with real-time data persistence and interactive AI-powered insights

Live Demo:

https://c1ye938tnkg0-deploy.space.z.ai/

WHAT'S NEXT

Future Roadmap

Phase 1

Real-time RSS News

Continuous market intelligence integration

Phase 2

PostgreSQL Migration

Multi-user production deployment

Phase 3

Redis Caching

Reduce AI API costs by 60-80%

Phase 4

Mobile Companion App

Push notifications for alerts

Thank You